

Test Cases for Java Chat Application

1. Server Startup

- Verify that the server starts successfully and listens on the correct port.
- Steps:
 1. Run the ChatServer application.
- Expected Result: Server starts without errors and displays "Chat server started..." message.

```
D:\Coding\Distributed System\Client-Server Chat>java ChatServer
Chat server started...
```

2. Multiple Client Connections

- Description: Ensure the server can handle multiple client connections simultaneously.
- Steps:
 1. Start the server.
 2. Connect 3 different client instances.
- Expected Result: Server accepts all connections and maintains them separately.

```
D:\Coding\Distributed System\Client-Server Chat>java ChatServer
Chat server started...
Client Connected
Client Connected
Client Connected
[]

D:\Coding\Distributed System\Client-Server Chat>java ChatClient
Enter a command (ECHO, ARITH, FILE) or type 'exit' to quit:
[]

D:\Coding\Distributed System\Client-Server Chat>java ChatClient
Enter a command (ECHO, ARITH, FILE) or type 'exit' to quit:
[]

D:\Coding\Distributed System\Client-Server Chat>java ChatClient
Enter a command (ECHO, ARITH, FILE) or type 'exit' to quit:
[]
```

3. Server Graceful Shutdown

- Description: Verify that the server shuts down gracefully, closing all client connections.
- Steps:

1. Start the server and connect multiple clients.
 2. Forcefully close the server (e.g., Ctrl+C).
- Expected Result: Server closes all client connections and shuts down without errors.

```

D:\Coding\Distributed System\Client-Server Chat>java ChatServer
Chat server started...
Client Connected
Client Connected
Client Connected
^C
D:\Coding\Distributed System\Client-Server Chat>

D:\Coding\Distributed System\Client-Server Chat>java ChatClient
Enter a command (ECHO, ARITH, FILE) or type 'exit' to quit:
Connection closed: Connection reset

D:\Coding\Distributed System\Client-Server Chat>java ChatClient
Enter a command (ECHO, ARITH, FILE) or type 'exit' to quit:
Connection closed: Connection reset

D:\Coding\Distributed System\Client-Server Chat>java ChatClient
Enter a command (ECHO, ARITH, FILE) or type 'exit' to quit:
Connection closed: Connection reset

```

4. Client Disconnection

- Description: Ensure that a client can disconnect gracefully.
- Steps:
 1. Connect a client to the server.
 2. Type 'exit' in the client console.
- Expected Result: Client disconnects from the server without errors.

```

D:\Coding\Distributed System\Client-Server Chat>java ChatServer
Chat server started...
Client Connected

D:\Coding\Distributed System\Client-Server Chat>java ChatClient
Enter a command (ECHO, ARITH, FILE) or type 'exit' to quit:
exit
Connection closed: Socket closed

D:\Coding\Distributed System\Client-Server Chat>

```

5. ECHO Command

- Description: Test the ECHO functionality.
- Steps:
 1. Connect a client to the server.
 2. Send command: `ECHO: Hello, World!`
- Expected Result: Client receives "Hello, World!" as a response.

```

D:\Coding\Distributed System\Client-Server Chat>java ChatServer
Chat server started...
Client Connected
Received: ECHO: Hello, World!

D:\Coding\Distributed System\Client-Server Chat>java ChatClient
Enter a command (ECHO, ARITH, FILE) or type 'exit' to quit:
ECHO: Hello, World!
Hello, World!
Enter a command (ECHO, ARITH, FILE) or type 'exit' to quit:

```

6. ARITH Command - Addition

- Description: Test basic arithmetic addition.
- Steps:
 1. Connect a client to the server.
 2. Send command: `ARITH: 5 + 3`
- Expected Result: Client receives "ARITH RESULT: 8" as a response.

```
D:\Coding\Distributed System\Client-Server Chat>java ChatServer
Chat server started...
Client Connected
Received: ARITH: 5 + 3
█

D:\Coding\Distributed System\Client-Server Chat>java ChatClient
Enter a command (ECHO, ARITH, FILE) or type 'exit' to quit:
ARITH: 5 + 3
ARITH RESULT: 8.0
Enter a command (ECHO, ARITH, FILE) or type 'exit' to quit:
█
```

7. ARITH Command - Division by Zero

- Description: Test error handling for division by zero.
- Steps:
 1. Connect a client to the server.
 2. Send command: `ARITH: 5 / 0`
- Expected Result: Client receives an error message about division by zero.

```
D:\Coding\Distributed System\Client-Server Chat>java ChatServer
Chat server started...
Client Connected
Received: ARITH: 5 / 0
█

D:\Coding\Distributed System\Client-Server Chat>java ChatClient
Enter a command (ECHO, ARITH, FILE) or type 'exit' to quit:
ARITH: 5 / 0
ARITH RESULT: ERROR: Division by zero
Enter a command (ECHO, ARITH, FILE) or type 'exit' to quit:
█
```

8. ARITH Command - Invalid Format

- Description: Test error handling for invalid arithmetic format.
- Steps:
 1. Connect a client to the server.
 2. Send command: `ARITH: 5 + + 3`
- Expected Result: Client receives an error message about invalid arithmetic format.

<pre>D:\Coding\Distributed System\Client-Server Chat>java ChatServer Chat server started... Client Connected Received: ARITH: 5 + + 3 █</pre>	<pre>D:\Coding\Distributed System\Client-Server Chat>java ChatClient Enter a command (ECHO, ARITH, FILE) or type 'exit' to quit: ARITH: 5 + + 3 ARITH RESULT: ERROR: Invalid arithmetic format Enter a command (ECHO, ARITH, FILE) or type 'exit' to quit:</pre>
--	---

9. FILE Command - Existing File

- Description: Test reading an existing file.
- Steps:
 1. Create a test file named "test.txt" with some content in the server's directory.
 2. Connect a client to the server.
 3. Send command: FILE: test.txt
- Expected Result: Client receives the contents of the test.txt file.

<pre>D:\Coding\Distributed System\Client-Server Chat>java ChatServer Chat server started... Client Connected Received: FILE: Test.txt █</pre>	<pre>D:\Coding\Distributed System\Client-Server Chat>java ChatClient Enter a command (ECHO, ARITH, FILE) or type 'exit' to quit: FILE: Test.txt Hello. How are you? Enter a command (ECHO, ARITH, FILE) or type 'exit' to quit:</pre>
--	--

10. FILE Command - Non-existent File

- Description: Test error handling for a non-existent file.
- Steps:
 1. Connect a client to the server.
 2. Send command: `FILE: nonexistent.txt`
- Expected Result: Client receives an error message about the file not being found.

<pre>D:\Coding\Distributed System\Client-Server Chat>java ChatServer Chat server started... Client Connected Received: FILE: Test1.txt █</pre>	<pre>D:\Coding\Distributed System\Client-Server Chat>java ChatClient Enter a command (ECHO, ARITH, FILE) or type 'exit' to quit: FILE: Test1.txt ERROR: Unable to read file - Test1.txt (The system cannot find the file specified) Enter a command (ECHO, ARITH, FILE) or type 'exit' to quit: █</pre>
---	--